

Naful Khalid

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EDUCATION

Florida International University (Honors College)

Miami, FL

- Computer Science (B.S.) (Dean's List) (Focus: Quantum Machine Learning)
- **Courses:** Data Structures, Algorithms Techniques, DBMS, Dynamic Programming, Computer Architecture

May 2027

TECHNICAL SKILLS

- **Languages:** Java, Python, C, C++, Go, Swift, JavaScript (ES6+), TypeScript, SQL, Assembly, HTML, CSS
- **Frameworks & Tools:** React, Node.js, Express.js, MongoDB, PostgreSQL, Bootstrap, Tailwind CSS, REST APIs
- **Cloud & DevOps:** IBM Z, Agile, Docker, Git, GitHub, UNIX, CI/CD, Software Docs

PROJECTS

AERO | Python, JavaScript, LLMs (Grok, Gemini), OpenCV, Selenium, DeepGram, 11Labs, OAuth

[GitHub](#) • [Demo](#)

- At TreeHacks-2025 (Stanford), developed a hands-free virtual assistant in 36 hours to execute voice commands
- Built speech-to-action processing with *DeepGram* & *11Labs* for real-time voice interactions with 95% accuracy
- Integrated *Gemini 1.5 Flash (Samara)*, *Gemini Vision*, *Selenium*, & *OpenCV* to extract on-screen content, *Grog LPU* for faster interpretation, reducing manual navigation time by 40%

CanaryOS | Go, Python, Pytorch, HTMX, PostgreSQL

[GitHub](#)

- Built a real-time voice-to-speech translation engine, reducing latency by 15% than Mozilla DeepSpeech
- With a team of 20, developed a NLP pipeline using test-driven development for streaming & meetings, improving translation accuracy by 3% and processing speed from 2 WPS to 4 WPS

Genie | React, JavaScript, TypeScript, WXT, Python, CSS, HTML

[GitHub](#)

- At HackNYU-2025, created a fintech browser extension, analyzing spending patterns & suggesting savings strategies in real-time with smart coupon tracking & cashback automation, raising user savings by 12%

HackEnzyme | JavaScript, HTML, CSS, Bootstrap, 3Dmol.js, Python, Jupyter Notebook

[GitHub](#)

- At HackHarvard-2024, created a bioinformatics tool predicting enzyme sequences from SMILES inputs with 3Dmol.js-based enzyme visualization & 85% prediction accuracy that garnered 30,000+ LinkedIn impressions

EXPERIENCE

Quantum Computing Research Intern (Miami, FL)

May 2025 - Present

- Researching complex-frequency (CF) pulse shaping in photonic quantum circuits to improve quantum gate fidelity, developing & benchmarking a custom simulation library to compare waveforms for non-Hermitian quantum control
- Working with IBM Quantum Qiskit Global Summer School, co-authoring a book & a research paper on CF pulse.

Technical Team Lead (INIT FIU - the largest tech student organization on campus)

Jan 2025 - April 2025

- Led a 15-member team to build a full-stack virtual marketplace for FIU community for digital & creative assets using MERN stack & CI tools, with an initial target user base of 500+ students
- Implemented agile workflow & improved development velocity by 25% than week 1 & reduced merge conflicts to 0

Teaching Intern (Computational Thinking, CIS1940)

Jan 2025 - April 2025

- Taught Computational Thinking (CT), algorithms, & Python to a class of 21 students at W.R. Thomas Middle School through INIT Ignite's outreach program, improving problem-solving quiz scores from 17% to 65% over the semester

Research Assistant Intern (Hybrid)

Jun 2020 - Nov 2024

- Published 2 Research papers & presented in 4 scientific conferences

AWARDS

- LinkedIn Top [Software Development](#) Voice (<3% quality contributors globally)
- Future of Florida Summit-2024 (UF) (Florida's largest student-run policy conference) ([2nd Runner Up Team](#))